

Cameron DeBough

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EDUCATION

University of Arizona, B.S. Computer Science & Artificial Intelligence, Minor in Mathematics

Expected May 2029

GPA: 3.87 · Dean's List with Distinction (Fall 2025) · Dean's List (Spring 2026) · 67 Credits Completed

Coursework: Software Development, Discrete Mathematics for CS I-II, Analyzing Data, Web Programming, Calculus I-II, Linear Algebra

EXPERIENCE

Software Engineering Intern, SkillfullyAware / SAAQ

October 2025 - Present

- Own the production backend of an AI-powered leadership assessment platform used by executive peer organizations (EO, YPO): Python and FastAPI services on Supabase (PostgreSQL), live in production
- Built a streaming LLM analysis pipeline with multi-stage scoring and two-pass generation that eliminated timeouts and truncated output, plus automated DOCX report generation from the analysis results
- Implementing authentication, Stripe payments, and persistent user storage for MVP launch

Founder & Engineer, DeBough Systems (deboughsystems.com)

August 2026 - Present

- Built and launched a security monitoring product for small businesses without IT staff: Next.js and TypeScript app with Supabase auth and Postgres, Stripe billing, and automated weekly re-scans
- Engineered a passive scanning engine that checks email security (SPF, DMARC, DKIM), TLS certificates, domain registration (RDAP), and registered lookalike domains, with transparent scoring and plain-English fix steps
- Hardened the app with a strict nonce-based CSP, security headers, SSRF-safe outbound requests, and rate limiting; required DNS ownership verification before breach monitoring so the tool can't be used for reconnaissance

Research Software Engineer, NewFOS Center, University of Arizona

March 2026 - Present

New Frontiers of Sound, an NSF Science & Technology Center

- Built Python automation pipelines for nonlinear acoustics experiments (hysteresis sweeps, harmonic SNR analysis, transfer-function extraction) used to model signal distortion in acoustic transducers
- Designed and validated an electrical impedance measurement rig with a full de-embedding pipeline that isolated a 42% cable-parasitic error in a previously reported measurement
- Developed MATLAB signal-analysis pipelines (SVD, dynamic mode decomposition) that produced publication-quality figures for experiments on a three-rod coupled acoustic waveguide
- Automated Tektronix oscilloscopes and function generators via VISA/SCPI for synchronized multi-channel data acquisition

Undergraduate Research Assistant, VSI Lab, University of Arizona ECE

September 2025 - Present

Embodied AI & Accessibility · Advisor: Prof. Eungjoo Lee

- Building 3D reconstruction (Gaussian splatting) and NVIDIA Isaac Sim environments for robot inspection, with ROS 2 control of LeRobot SO-101 arms and TurtleBot4 navigation
- Building computer-vision pipelines for Meta Aria 2 smart glasses (hand tracking, gaze data, real-time on-device ML inference) for ASL accessibility and skill-assessment applications

TECHNICAL PROJECTS

APEX QUANT, Algorithmic Futures Trading System

- Architected event-driven backtesting and paper-trading engine for micro futures (MES/MNQ/MGC) with modular strategy framework, risk management controls, and a broker-integration layer targeting Interactive Brokers (ib_insync)

Thermal-Tracking Turret, Embedded Systems / Robotics

- Built an autonomous heat-tracking pan/tilt turret on Raspberry Pi: AMG8833 8x8 thermal array over I2C, PCA9685 hardware PWM servo control, and a closed-loop feedback algorithm that centers a moving heat source in real time, deployed as a headless systemd service

Synapse Finance, Full-Stack Financial SaaS Platform

- Shipped production app with 50+ active users: Flask, React, PostgreSQL, JWT authentication, RESTful APIs, and CI/CD

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, SQL, MATLAB, Bash, HTML/CSS

Frameworks: FastAPI, Flask, React, Next.js, Node.js, Tailwind CSS

Embedded & Hardware: Raspberry Pi, I2C sensors, PWM servo control, VISA/SCPI instrument automation, ROS 2

Data & ML: NumPy, SciPy, Pandas, signal processing (FFT, SVD), computer vision, LLM integration

Tools & Databases: Git, Docker, Linux, PostgreSQL, Supabase, Stripe, CI/CD

LEADERSHIP & ACTIVITIES

Coke Fellow, The Coca-Cola Company, Introships Career Discovery Program

May - July 2026

- Selected as 1 of 50 fellows nationwide for a coached cohort program with Coca-Cola executives

University of Arizona AI Club · MLH Global Hack Week · 13+ GitHub projects